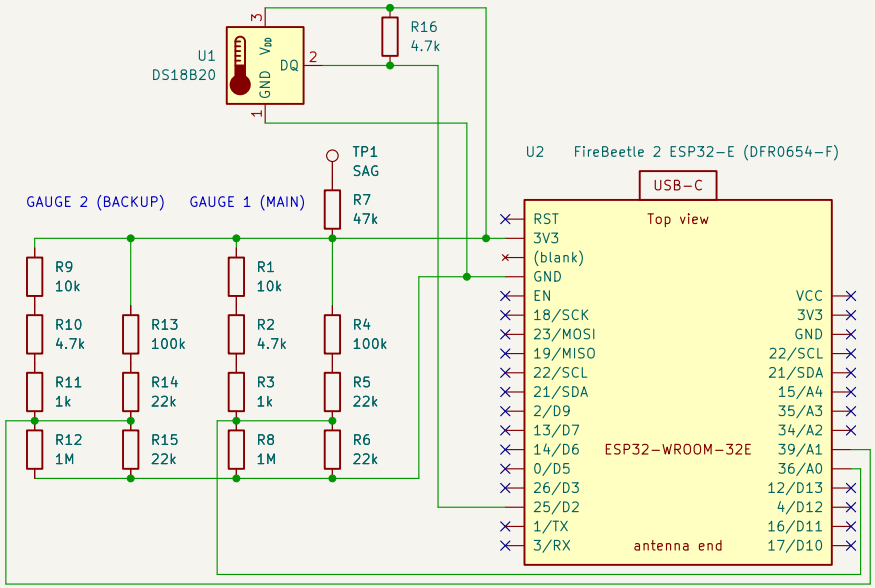


NOTES
Gauge 1 = MAIN supply (R1–R8) -> A0 / GPIO36. Gauge 2 = BACKUP supply (R9–R15) -> A1 / GPIO39.
Each gauge: top 10k+4.7k+1k from the supply, bias 100k+22k from 3V3, bottom 22k to GND, 1M pull-down.
DS18B20 (U1) on D2 / GPIO25 with a 4.7k pull-up (R16). The sensor cable shield is not connected.
Breadboard tests, measured at the pin against GND (gauge 1 / gauge 2):
  healthy: as drawn -> 1.98 V / 1.98 V
  sag: R1 supply wire moved from 3V3 to TP1 (SAG), so the supply comes through R7 47k -> 1.10 V (gauge 1 only)
  dead: supply wire unplugged, the 100k+22k bias keeps a live zero -> 0.488 V / 0.489 V
  broken wire: wire from the gauge to the ESP32 pin unplugged -> 0 V / 0 V
  (on the breadboard the 1M sits at the pin end of that wire, so a broken wire reads 0 V, not a floating value)
Power: USB only. Nothing above 3.3 V on any pin. No mains anywhere.

USB power only. Nothing above 3.3 V on any pin. No mains.
Gauge live zero: dead supply 0.49 V, healthy 1.98 V, broken wire 0 V
Served to the PLC as a Modbus TCP server over Wi-Fi (code: Part 1, phase C)
FireBeetle 2 ESP32-E: two supply gauges (A0, A1) and a DS18B20 (D2)
Abdi – github.com/justkuchkorov

Sheet: /
File: transformer-point.kicad_sch

Title: Transformer point – field level (Part 1)

Size: A4	Date: 2026-11-04	**Rev: 1**
KiCad E.D.A. 10.0.5		Id: 1/1